Individual Reflections

Linnea:

My main part of the project was verifying the AI-built master dataset against the raw datasheets, and then combining my checks into the final script. Rather than trusting the AI's own verification table, I traced every row of the master back to its original Excel cell and tested each QC flag, exclusion and name mapping on its own. AI was useful for writing the R code quickly and for these complex data cleaning steps and for walking me through Git merge conflicts, which happened constantly, but AI was wrong multiple times with its code. Its first comparison reported 370 "mismatches" that were just a bug in its own code, and at one point it concluded that the master had put two surveys on the wrong island. Which is why I constantly checked against the raw Excel files to personally verify the claims and assumptions AI gave me: missing site names, and classifying a fish as an invertebrate. I'd trust AI-generated code when its output can be checked against something concrete, like a row count or a raw cell, especially when working with a dataset I'm not familiar with. I wouldn't trust it for judgment calls, such as which site is correct or how to classify a species, without checking the source myself, because AI sounded just as confident when it was wrong. I also completed the final verification of the final script to fix errors in order to run and render once we combined all of our individual scripts into one, and clean up some repetitive directory calls and library loading at different places throughout the script, as well as cleaned up the GitHub repository.

Carolina:

For our group project, my main contributions were creating the functions from our master datasheet, writing the README, and putting together our final Quarto script with everyone's code. I had a positive experience using AI to help me create the functions because I was able to explain what I wanted each function to do, and AI helped me figure out how to write the code. I felt like I was able to understand the functions that were created, and I was able to test them using our cleaned data and it worked perfectly. The most challenging part of the project was combining everyone's code into one final script. Since all four of us used AI to help with our individual scripts, we ended up with different coding styles even when we were working with the same dataset. I also noticed that some of the AI generated code was super different from how we have learned in class this semester. This made it hard to combine everything into one final script and get it to run smoothly. I ran into several errors, especially when trying to render the final

script. AI was helpful with troubleshooting, but there were times where the suggestions were not working and it felt like it made things much more complicated than they needed to be. Overall, this experience taught me that AI is super helpful, but obviously we should not trust everything it suggests.

Jasmine:

For this assignment I generated various plots and a plot function with the help of AI. Our group specifically wanted to focus on the algae data collected from the Opihi project, so I first began with cleaning data. From the newly organized master sheet, I used AI to help me generate new data that only included algae observations, as well as classify those algae as either native or non-native. AI gave me a data sheet that re-organized all limu species into either "Native", "Non-native", or "Unclassified", and in the last column, provided the source where it found this information. This was iffy, I double checked all the classifications with *Hawaiian Reef Plants,* but I'm still not convinced that that is enough verification. Community members should be providing those classifications, and I think the use of AI for native species is a touchy subject all around. I then generated three interactive map plots, starting with what we learned in class and asked AI to plot the data on the leaflet, which I thought it did very well and saved me a lot of time. Lastly, I generated a plot function to display the change of percent cover of native and non-native over time for each site. AI did a good job, but added lots of unnecessary layers that made the plots look busy when faceting them together.

Cameron:

My role in our group project was primarily in the original data cleaning. I found AI to be incredibly useful in helping me conceptualize different methods for cleaning the data, as I was able to upload all of our redacted datasheets to AI and have dialogue on goals and steps to take. I probably had 30 minutes of back and forth discussion with Claude before any coding actually happened, and by that point we had a very solid plan and Claude was great at assisting with the workflow. Some bugs came up now and then, and some things did not work perfectly, but having the long initial conversation really helped Claude resolve these issues swiftly. I went through and examined the outputs, but additionally, I also had Claude create checks and different verifications

to ensure things were working smoothly. I think the biggest limitation is that we did not have direct access to the data owner to ask questions, so occasionally Claude would have a question for me that I could not be 100% certain of the answer, but I did the best that I could. I think that leads me to when I would/would not trust AI generated code. If the data is mine, I understand every part of the raw data, I have a very clear understanding of the research question, the process from collection to analysis is clear, and I can understand the majority of the generated code, I have little problem with trusting AI-generated code.